Mahmoud Alyosify

Academic CV | PhD Applicant in Artificial Intelligence
Efficient Language-Model Inference · Trustworthy & Evaluable AI Systems
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RESEARCH INTERESTS

Efficient inference for large language models — training-free decoding-time methods, computational and energy cost of generation · **Rigorous evaluation of generative systems** — non-inferiority frameworks, length-controlled judging, reproducibility · **Self-supervised and contrastive representation learning** · **Reinforcement learning for resource-constrained decision-making** · **Multi-agent systems and knowledge-graph-grounded reasoning** · **Empirical software engineering in the age of coding agents**.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation* — Supervisor: **Dr. Rahatara Ferdousi**.
- Graduate coursework: **Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End Machine Learning Pipelines**.

B.Sc. Computer & Information Science — Bioinformatics — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Undergraduate thesis: **Vitalism Solution** — contactless vital-sign estimation from video using remote photoplethysmography (rPPG) and wavelet-based signal filtering. **Grade: A+**.
- **Semi-Finalist, Microsoft Imagine Cup 2023**; **Finalist, ISEIC 2023** (international research competition).

RESEARCH EXPERIENCE & PROJECTS

MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation — *2026 – Present*
Master's Research Project (CISC 898), Queen's University · Supervisor: Dr. Rahatara Ferdousi · code

- **Research question:** can a training-free controller inside the decoding loop cut LLM output tokens while preserving content, measured against the simplest baseline — truncating the model's own answer to the same length? Tested on **Mistral-7B, Llama-3.1-8B and Qwen2.5-3B**.
- **Finding:** an entropy-gated **logits processor** removed **25.5–34.9%** of tokens on MT-Bench development questions (**19.9–34.2%** on a sealed test split) within pre-registered BERTScore margins, but an audit showed every output was a prefix of the baseline: the saving was truncation. A survey of **23** training-free mechanism families found none that beats truncation without oracle labels.
- **Method: Entailment-Licensed In-Decoding Elision** — a sentence encoder proposes spans close to the text already shown, an NLI model licenses a span only if that text entails it. It beats truncation on every model while removing 0.9–2.2% of tokens.
- **Human evaluation:** on 670 public prompts readers judged 49–69% of elisions harmful; a version scoped to its discourse unit lowered this to **10.2%** (95% CI 7.5–13.7%) on two fresh tests. Behaviour transfers to sampling, batching, other precisions, six closed systems and six languages.
- **Conclusion:** ordinary answers contain little genuine restatement; larger cuts discard content users asked for. Runs are deterministic, with fixed seeds, versioned environments and a pinned **NVIDIA A40** GPU.

Analyzing Review Effort in Human vs. Agentic Pull Requests — *2026*
Empirical Software Engineering · Repository Mining · code

- Built a reproducible mining pipeline over large-scale open-source repositories to quantify whether AI-agent-generated pull requests shift code-review effort relative to human-authored contributions, with explicit cohort definitions and inclusion criteria.

Self-Supervised Contrastive Representation Learning — A Controlled Augmentation Study — *2026*
SimCLR · SupCon · PyTorch · code

- Conducted a **42-experiment augmentation sweep** isolating which transformations drive representation quality, extended with a semi-supervised **SupCon** objective; **84.30% top-1 accuracy** with retrieval evaluation via ONNX/FAISS.

Deep Reinforcement Learning for Cloud Resource Provisioning — *2026*
PPO · DQN · Custom Gymnasium Environment · code

- Designed a custom simulation environment and multi-objective **cost–latency reward**; compared PPO and DQN against static and threshold-based baselines representative of deployed practice.

HORUS Sentinel — Multi-Agent Intelligence Synthesis over a Knowledge Graph *2026*
Multi-Agent Systems · Fine-Tuned LLM · Knowledge Graphs · code

- Architected a swarm of specialised autonomous agents building a **living knowledge graph** as shared memory, over which a self-hosted fine-tuned model produces prioritised, source-traceable reports.

Further Research Output

- **Horus-OSINT** — domain adaptation of **Meta-Llama-3-8B** over **159,826** GTD/GDELT records, deployed on AWS.
- **SCRAP** — predictive modelling of satellite collision risk constrained to a **48-hour** operational lead time.
- **AudioShield** — CNN-based synthetic-speech detection over mel-spectrogram representations.
- **Open research artefacts:** GitHub · Hugging Face · Kaggle · Portfolio.

TEACHING EXPERIENCE

Part-Time Instructor — Machine Learning & AI for Business *Jul 2025 – Present*
National Telecommunication Institute (NTI) Cairo, Arish & Assiut, Egypt

- Teach **Machine Learning** (90-hour bootcamps in Cairo, Arish and Assiut) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

Online Instructor — Mathematics, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Authored Arabic-language curricula in **discrete mathematics, algorithms, probability, statistics** and **graph data mining**, reaching **10,000+ learners** and **700K+ views**.

PROFESSIONAL EXPERIENCE

AI Engineer Intern *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) Maadi, Cairo, Egypt

- Building an agentic RAG assistant (LangChain, LangGraph, FAISS, FastAPI) with measurement-driven design: retrieval benchmarks (top-1 accuracy **75% to 94%**), evaluation of LLM graders on labelled pairs, and LLM-judge reliability checked against deterministic metrics (0.82 agreement).

AI in Cybersecurity Intern *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) Cairo, Egypt

- Applied AI methods across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, Entity Framework, SQL Server**.

RESEARCH & TECHNICAL SKILLS

Methodology	Pre-Registered Protocols, Paired Non-Inferiority Testing, Hypothesis Testing, Ablation & Sweep Design, Baseline Construction, Reproducible & Deterministic Experimentation, Scientific Writing
Frameworks	PyTorch, Hugging Face Transformers / PEFT / Datasets, TensorFlow, Scikit-learn, Gymnasium, Optuna, SHAP, NumPy, Pandas, PySpark, FAISS, ONNX, LangChain, LangGraph
Infrastructure	NVIDIA A40 GPU, AWS, Docker, FastAPI, CI/CD, Git/GitHub, Linux, NVML Energy Profiling
Programming	Python, C++, C, SQL, C#, Go, R, MATLAB, Bash

AWARDS, SCHOLARSHIPS & SERVICE

- **Digilians Presidential Scholarship**, Queen's University — competitive national AI graduate fellowship (2025).
- **Semi-Finalist**, Microsoft Imagine Cup 2023 · **Finalist**, ISEIC 2023 · **1st Place**, Smart Cities Hackathon 2022 · **1st Place**, Science & Technology Content Competition.
- **Ideal Student Award**, Faculty of Computers & Information, Assiut University, 2022–2023.
- **Service:** Secretary, Higher Scientific & Technological Committee, Student Union, Assiut University · Ambassador, Information Technology Institute (ITI), Assiut · Volunteer Team Leader, USAID University Center for Career Development.
- **Languages:** Arabic (Native) · English (Professional Working Proficiency) · French (Basic). **References available on request.**